

1. Write MATLAB expressions to print your name and registration number.

[2 marks]

2. Using MATLAB

- a. Define a vector t to contain elements from 0 to 20 in steps of 0.1
- b. Given that the solution of a DE is $y = R - R * \exp(-0.5 * t)$ where R represents the last two digits of your registration number
 - i. Plot y from 0 to 20.
 - ii. Label the x and y axes appropriately.
 - iii. Set the title of the Figure to your registration number.
 - iv. Limit the y axis appropriately so the plot can be viewed properly.

[9 marks]

3. A system has a forward loop transfer function

$$G(s) = \frac{K}{s^4 + 12s^3 + 64s^2 + 128s}$$

Using MATLAB

- a. Show the gain margin (GM) and phase margin (PM) of the system and the associated frequencies on a Bode plot.
- b. Find an expression for the closed loop transfer function G_{cl} with the gain being set at $R \times 7$ where R is the last two digits of your registration number.
- c. Find the poles of the system in (b), show its step response and comment on its stability.
- d. On the same Figure, show the step response of the system with the gain set at $GM - R$.

[9 marks]